VINAY KUMAR

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EDUCATION

Indian Institute of Information Technology, Una **Aug 2023 – May 2027**
B.Tech in Electronics and Communication Engineering *CGPA: 7.25*

EXPERIENCE

Full Stack Developer Intern — Sahii **Remote | Present**
Wholesale e-commerce platform

- Engineered and shipped **15+ production REST APIs** and server-side business logic in **Node.js/Express/MongoDB**, powering core catalog, cart, and order workflows for a live wholesale e-commerce platform.
- Integrated **Prisma ORM** with MongoDB for type-safe database access and **schema/data modeling**, cutting query-development time by **~30%** and reducing runtime data errors by **~25%**.
- Built typo-tolerant, relevance-ranked product search using **Meilisearch and Fuse.js**, and optimized MongoDB queries, reducing average search response time by **~40%** and improving result relevance.
- Automated procurement workflows via a **Vendor Management System (VMS)**, using a **message queue** to sync orders in real time between the e-commerce platform and the VMS, cutting manual order-entry effort by **~50%**.

PROJECTS

Mendly — Electronics Repair Service Platform **Live | GitHub**
React, Node.js, Express.js, MongoDB, Socket.IO, Razorpay, RAG, Pinecone

- Built a full-stack repair-service marketplace connecting customers with verified providers across **3 RBAC roles**, featuring booking management, live chat, order tracking, and reviews.
- Implemented **JWT access/refresh authentication** with httpOnly cookie sessions, Redux state management, and **rate limiting** (100 req/15 min), plus **Socket.IO** for real-time messaging and **Razorpay** for secure payments.
- Structured the backend using an **MVC pattern** with MongoDB indexing and **Zod** validation; deployed on **Azure Web App**, achieving **sub-150ms average API response time**.
- Integrated a **RAG-powered support chatbot** using the **Groq API and Pinecone vector database**, retrieving relevant context to deliver low-latency, accurate responses to customer queries.

Chaos Proxy — HTTP Fault-Injection Proxy **GitHub**
Node.js, TypeScript, Express, React, Socket.IO, WebSockets

- Built an HTTP fault-injection proxy from scratch on raw Node.js HTTP streams, handling concurrent connection forwarding, piping, **asynchronous I/O**, and error recovery without a framework.
- Designed a **triangle-wave fault degradation model** to simulate realistic network failure patterns, with a real-time pipeline computing **p50/p95/p99 latency and failure-rate metrics**.
- Built a **threshold-based assertion engine** that auto-halts fault injection on p99 latency or failure-rate breaches, configurable live via **REST APIs**; shipped a live **React and Socket.IO dashboard** visualizing telemetry and fault trends with **Recharts**.

DocFusion — AI Document Analysis Platform **Live | GitHub**
FastAPI, React, OCR, RAG, FAISS, MiniLM, Groq API, Redis, RQ, Supabase, SlowAPI, Docker

- Developed an AI-powered document analysis platform using **FastAPI, OCR, and a Retrieval-Augmented Generation (RAG) pipeline** to analyze PDFs, scanned images, and text files, generating summaries for **50+ page documents in under 10 seconds**.
- Built a **FAISS-based semantic retrieval pipeline** with Sentence Transformer (MiniLM) embeddings and Redis/RQ background workers, integrating the **Groq API (Llama 3.3)** for low-latency AI inference.
- Integrated **Supabase authentication, SlowAPI rate limiting**, and document history with a **Dockerized deployment**, optimizing the end-to-end OCR, retrieval, queue, and LLM pipeline for consistently fast processing.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL, HTML, CSS
Backend & Databases: Node.js, Express.js, REST APIs, PostgreSQL, Prisma, MongoDB, Redis, Supabase
Frameworks & Libraries: React, Next.js, Socket.IO, Redux, Tailwind CSS, Bootstrap, Zod
AI / LLM & RAG Systems: RAG, Vector Databases (Pinecone, FAISS), Semantic Search, Groq API, Llama 3.3, LangChain
Developer Tools & DevOps: Git, GitHub, Postman, Docker, VS Code, NPM, AWS EC2, Azure Web App, GitHub Actions, Vercel, Render
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS

- **Solved 500+ DSA problems** across LeetCode, CodeChef, and GeeksforGeeks; achieved a **2-Star rating** on CodeChef.
- **Published 2 open-source npm packages**, including Auth7-kit, a reusable authentication utility library.
- **Secured 3rd position among 200+ teams** at Hack the Hills 2025 for a working full-stack prototype.
- **Social Media Head, EPMOC Club, IIIT Una** — led social strategy and promotional content for technical events.